

Bhavana Sudhakar Athavane

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Education

California State University, Fullerton

Master of Science, Computer Science

Fullerton, CA

May 2026

Jyothy Institute of Technology

Bachelor of Engineering, Computer Science

Bangalore, India

July 2022

Experience

California State University

Graduate Teaching Assistant

Fullerton, CA

January 2026 – August 2026

- Instructed SQL and applied analytics to 50+ students/semester; redesigned lab exercises around joins, aggregations, window functions, query debugging, and techniques for handling large datasets (100K+ rows) in Microsoft Excel

Nokia

Sunnyvale, CA

Platform Engineering Architect → Database Software Engineer (Co-Op)

May 2025 – December 2025

- Built an incident-driven root-cause diagnostic PoC on Google Cloud Vertex AI, combining generative models with retrieval augmented (RAG) search over historical incident data; tuned prompts and validated recommendations against real Jira incidents, achieving ~82% root-cause match accuracy and cutting MTTR by ~41%
- Architected a GCP data pipeline (Pub/Sub → Dataflow → BigQuery) to ingest and transform incident log bundles and diagnostic data from 200+ network nodes; reduced manual triage effort by ~80% and enabled automated alerting
- Scaled the retrieval layer (RAG) using Vertex AI Search (Discovery Engine), unifying CMDB, Jira, and distributed knowledge bases into a single cross-source semantic search backend; improved runbook relevance and reduced incident resolution time by ~35%
- Designed and optimized 4+ relational database schemas across Oracle, MySQL, and PostgreSQL underpinning the platform's AI-driven backend systems; reduced average query latency by ~55% through query-plan analysis, SQL rewrites, and indexing
- Built Flask-based REST API integrations connecting backend services to OLTP-like data sources (Jira, CMDB), standardizing data access and reducing redundant database queries

Cognizant

Bangalore, India

Senior Systems Engineer

March 2022 – August 2024

- Led an Oracle DB performance engineering initiative across 3 production systems, improving overall throughput by 30% through query-plan analysis, index rebuilding, and partition pruning; directly supporting SLA compliance
- Designed and scaled an automated SQL data-validation framework in Python and PL/SQL spanning Oracle DB and SQL Server across 4+ production pipelines; caught ~20 cross-system reporting discrepancies per month
- Built and delivered 6+ executive Tableau dashboards tracking KPIs across revenue, operations, and compliance, enabling data-driven decision-making for senior stakeholders
- Automated over 8+ internal ETL workflows using Python and SQL, reducing end-to-end processing time by ~15%

Technical Skills

- Languages & Platforms:** Python, SQL, Java, JavaScript, Git, Tableau, Jira, REST APIs, Flask
- Cloud & AI/ML:** GCP, Vertex AI, BigQuery ML, Cloud Storage, AWS Redshift, LLM Integration, RAG, Semantic Search, NLP
- Data Engineering:** ETL Pipelines, Database Design, Query Optimization, Stored Procedures, Pub/Sub, Dataflow
- Databases:** Oracle DB, MySQL, SQL Server, PostgreSQL

Projects

AI-Powered Financial Fraud Detection System

- Built an end-to-end financial fraud detection pipeline covering transaction ingestion, feature engineering, and anomaly classification; model identified high-risk patterns across a dataset of 1M+ synthetic transactions with 89% precision
- Engineered a feature set from transaction metadata (velocity, geolocation deltas, merchant category patterns) to improve model separability between fraudulent and legitimate transactions
- Evaluated model performance using precision-recall tradeoffs suited to class-imbalanced fraud data, tuning classification thresholds to minimize false positives while preserving recall on rare fraud cases

Content Modeling using NLP

- Developed an NLP document processing pipeline using CNN models and OCR to extract and structure unstructured text from 5+ document formats, reducing manual categorization effort for a 10K-document test corpus
- Built a preprocessing layer for OCR noise correction and layout normalization, improving downstream text extraction accuracy across scanned and image-based document formats
- Designed a document classification scheme to auto-tag extracted content by category, structuring previously unstructured data for downstream search and retrieval